

Quick Start Guide - Mental Health App

5-Minute Setup

Step 1: Install Dependencies (1 minute)

```
pip install -r requirements.txt
```

Step 2: Run the App (30 seconds)

```
streamlit run mental_health_app.py
```

Step 3: Open in Browser

The app will automatically open at:

```
http://localhost:8501
```

What You'll See

The app has 4 main sections (use sidebar to navigate):

Overview

- Dataset statistics
- Project objectives
- Key metrics

EDA & Visualizations

- Global depression trends
- Top countries ranking
- Disorder correlations

- Statistical summaries

Model Building

- Model performance metrics
- Training results ($R^2 = 0.943$)
- Feature importance
- Actual vs Predicted visualization

Predictions

- **Interactive prediction interface**
 - Enter Year, Anxiety, Drug Use, Alcohol rates
 - Get instant depression rate prediction
 - View 20-year trend forecast
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Try This First

1. Go to  **Predictions** page
 2. Use the default values (2025, 3.8%, 0.95%, 1.4%)
 3. Click "  **Predict Depression Rate** "
 4. You should see: **Predicted Depression: ~3.847%**
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Dataset Info

- **6,468 records** from 200+ countries
 - **30 years** of data (1990–2019)
 - **7 mental disorders** tracked
 - Uses generated sample data (replace with your CSV)
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Troubleshooting

Port already in use?

```
streamlit run mental_health_app.py --server.port 8502
```

Need to install Python?

- Download from python.org
- Make sure to check "Add Python to PATH" during installation

On Mac/Linux permission issue?

```
chmod +x mental_health_app.py
```

Next Steps

1. Replace sample data with your actual CSV:

- Update `load_and_process_data()` function
- Use: `df = pd.read_csv('your_dataset.csv')`

2. Retrain model with your data

3. Deploy online (see README.md for options)

Quick Reference Commands

```
# Install all dependencies
pip install -r requirements.txt

# Run the app
streamlit run mental_health_app.py

# Run on custom port
streamlit run mental_health_app.py --server.port 8502

# Run in headless mode (no browser)
streamlit run mental_health_app.py --logger.level=debug

# Deactivate virtual environment
deactivate
```

✨ Features Highlight

✅ 4 interactive pages with navigation ✅ 94.3% accurate ML model ✅ Real-time predictions ✅ Beautiful visualizations (matplotlib + seaborn) ✅ Responsive design ✅ Fast data processing with caching

Ready to go! 🚀 Run the app and explore the global mental health insights!